

6 K-SPACE SAMPLING STRATEGIES

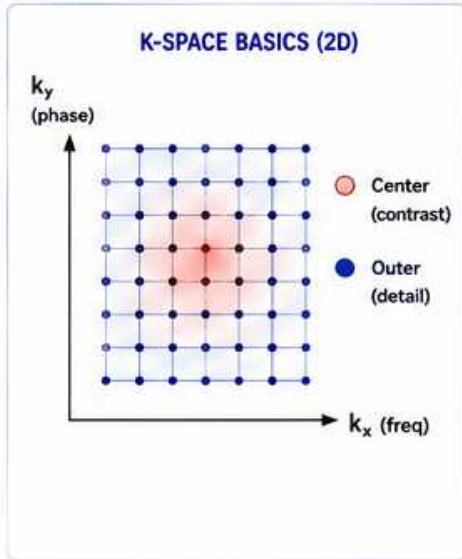
Different ways to fill k-space – each with trade-offs in speed, detail, and artifacts.

THE BIG IDEA

The way k-space is sampled determines scan time, image quality, and artifacts. There is no single “best” strategy – only the best choice for the clinical goal.

THE CONCEPT

- 1 K-space can be sampled in many different orders.
- 2 Center k-space (low frequencies) is most important for contrast.
- 3 Outer k-space (high frequencies) adds detail.
- 4 Different sampling patterns affect scan time, motion sensitivity, and artifacts.
- 5 Understanding these trade-offs helps choose the right strategy.



COMMON K-SPACE SAMPLING PATTERNS

1. LINEAR (SEQUENTIAL) Line by line	2. CENTRIC LINEAR Center-out (line order)	3. CENTRIC REVERSE LINEAR Reverse center-out	4. RADIAL Spokes	5. SPIRAL Continuous spiral	6. ELLIPTICAL / PROFILE ORDER Elliptical shells
• Lines acquired sequentially (from one side to the other) • Simple and easy to implement • Sensitive to motion	• Center of k-space acquired first, then outer lines • More robust to motion (for SE) • Same scan time as linear	• Outer k-space first, center acquired last • Useful for contrast changes during scan (e.g., CE timing)	• Lines pass through the center (like spokes) • Very motion robust • More time to fill center (less efficient)	• Continuous spiral path outward from center • Very fast acquisition • Sensitive to off-resonance and gradient imperfections	• Samples k-space in elliptical shells from center outward • Balance of efficiency and robustness
Used in: Spin Echo (SE), many 2D scans	Used in: SE, FSE (echo train centered)	Used in: Gradient Echo, Dynamic studies	Used in: fMRI, ASL, Trauma, Free-breathing	Used in: fMRI, MR Angiography (MRA), Fast imaging	Used in: Many 3D sequences, Balanced SSFP

HOW SAMPLING ORDER AFFECTS MOTION

MOTION EARLY (Linear)
Center acquired late

More artifacts (ghosting)

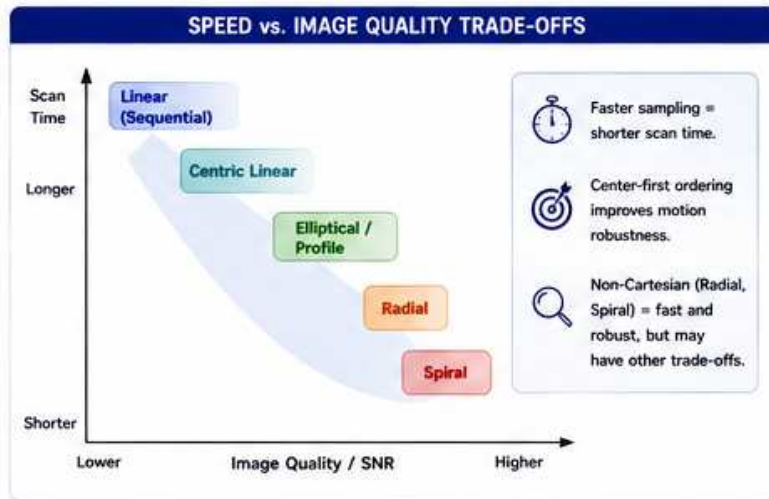
MOTION WITH CENTRIC
Center acquired first

Fewer artifacts (better image)

MOTION LATE (Reverse Centric)
Center acquired last

Contrast changes may affect image

In general: Centric ordering is more motion robust for Spin Echo.



NON-CARTESIAN vs. CARTESIAN

CARTESIAN (Linear, Centric, Elliptical)

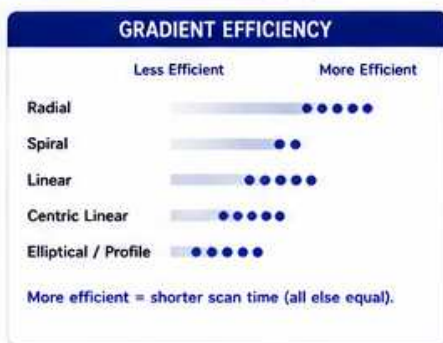
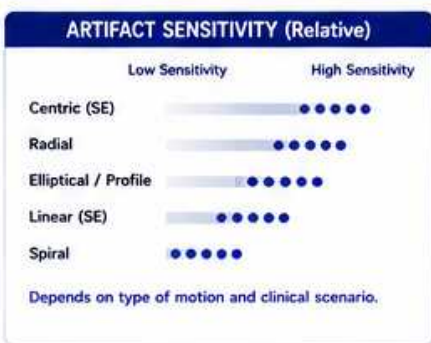
- Straight lines in k-space
- Easy to reconstruct
- More sensitive to motion (but centric helps)
- Efficient use of gradients

NON-CARTESIAN (Radial, Spiral)

- Curved paths in k-space
- Very motion robust
- Fast acquisition
- Reconstruction more complex
- Sensitive to off-resonance (spiral more than radial)

CLINICAL EXAMPLES

- Brain Routine (SE/FSE):** Centric Linear
- fMRI / ASL:** Radial or Spiral
- MRA (TOF/CE):** Spiral
- Trauma / Free-breathing:** Radial
- 3D Imaging (SSFP):** Elliptical / Profile



SUMMARY OF STRATEGIES

Strategy	Best For	Advantages	Limitations
Linear (Sequential)	General imaging	Simple, robust (no ordering complexity)	Motion sensitive
Centric Linear	SE / FSE	Motion robust, same scan time	Center lines may be most critical
Centric Reverse	Dynamic / Contrast	Center last (good for timing)	More motion sensitive
Radial	Motion, fMRI, Trauma	Very motion robust	Less efficient, longer reconstruction
Spiral	MRA, fMRI, Fast scans	Very fast	Off-resonance sensitive
Elliptical / Profile	3D, Balanced SSFP	Good balance	More complex than linear

KEY TAKEAWAYS

- ✓ Center k-space = contrast; Outer k-space = detail.
- ✓ Sampling order affects motion sensitivity and contrast timing.
- ✓ Centric ordering improves motion robustness in SE.
- ✓ Radial = most motion robust; Spiral = fastest but sensitive.
- ✓ Choose the strategy that best matches the clinical goal.